

Continuous Distributions	
Normal (or Gaussian) Distribution ✓	
Continuous Uniform Or Rectangular Distribution ✓	
Exponential Distribution ✓	
Gamma Distribution ✓	
Beta Distribution ✓	(2)
Cauchy Distribution ✓	
Weibull Distribution ✓	
Lognormal Distribution ✓	
Pareto Distribution ✓	

Rectangular Unifn Distribution

A r.v 'x' is said to have a continuous uniform distribution over the interval (a, b). The

p.d.f is defined as

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Question

Find the rth moment about origin of the uniform distribution also find the moments about means.

Sol $\mu'_r = \int_a^b x^r f(x) dx$

$$= \frac{1}{b-a} \int_a^b x^r dx$$

$$= \frac{1}{b-a} \left| \frac{x^{r+1}}{r+1} \right|_a^b$$

$$\mu'_r = \frac{1}{(b-a)(r+1)} [b^{r+1} - a^{r+1}]$$

$$\mu'_1 = \frac{1}{(b-a)(2)} (b^2 - a^2)$$

$$= \frac{(b-a)(b+a)}{(b-a)(2)} = \frac{a+b}{2}$$

$$\mu'_2 = \frac{1}{(b-a)(3)} (b^3 - a^3)$$

$$\mu_2' = \frac{(b-a)(a^2+ab+b^2)}{3(b-a)} \quad (2)$$

$$= \frac{a^2+ab+b^2}{3}$$

$$\mu_3' = \frac{1}{4(b-a)} (b^4-a^4) \quad r=3$$

$$= \frac{(b^2)^2 - (a^2)^2}{4(b-a)} = \frac{(b^2-a^2)(b^2+a^2)}{4(b-a)}$$

$$= \frac{(b-a)(b+a)(b^2+a^2)}{4(b-a)}$$

$$\mu_3' = \frac{(a+b)(a^2+b^2)}{4} \quad r=4$$

$$\mu_4' = \frac{1}{5(b-a)} (b^5-a^5)$$

$$= \frac{1}{5(b-a)} (b-a)(b^4+ab^3+a^2b^2+a^3b+a^4)$$

$$\mu_4' = \frac{b^4+ab^3+a^2b^2+a^3b+a^4}{5}$$

Moments about mean, 5

$$\mu_1 = 0$$

$$\mu_2 = \mu_2' - (\mu_1')^2$$

$$\mu_2 = \frac{a^2 + ab + b^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{a^2 + ab + b^2}{3} - \frac{a^2 + b^2 + 2ab}{4}$$

$$= \frac{4a^2 + 4ab + 4b^2 - 3a^2 - 3b^2 - 6ab}{12}$$

$$\mu_2 = \frac{a^2 + b^2 - 2ab}{12}$$

$$\mu_2 = \frac{(a-b)^2}{12}$$

$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2(\mu_1')^3$$

$$= \frac{(a+b)(a^2+b^2)}{4} - 3\left(\left(\frac{a^2+ab+b^2}{3}\right)\left(\frac{a+b}{2}\right)\right) + 2\left(\frac{a+b}{2}\right)^3$$

$$= \frac{a+b}{2} \left[\frac{a^2+b^2}{2} - (a^2+ab+b^2) + \frac{(a+b)^2}{2} \right]$$

$$= \frac{a+b}{2} \left[\frac{a^2+b^2}{2} - (a^2+ab+b^2) + \frac{a^2+b^2+2ab}{2} \right]$$

$$= \frac{a+b}{2} \left[\frac{a^2+b^2 - 2a^2 - 2ab - 2b^2 + a^2 + b^2 + 2ab}{2} \right]$$

$$\mu_3 = \frac{a+b}{2} (0)$$

$$\mu_3 = 0$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'(\mu_1')^2 - 3(\mu_1')^4$$

$$= \frac{b^4 + ab^3 + a^2b^2 + a^3b + a^4}{4} - 4\left[\frac{(a+b)(a^2+b^2)}{4}\left(\frac{a+b}{2}\right)\right] + 6\left[\left(\frac{a+b}{2}\right)^2\left(\frac{a^2+ab+b^2}{3}\right)\right] - 3\left[\left(\frac{a+b}{2}\right)^4\right]$$

$$= \frac{b^4 + ab^3 + a^2b^2 + a^3b + a^4}{5} - \frac{a^4 + 2a^2b^2 + 2a^3b + 2ab^3 + b^4}{2} \quad (4)$$

$$+ \frac{a^4 + 4a^2b^2 + 3a^3b + 3ab^3 + b^4}{2} - \frac{3}{16} (a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4)$$

$$= \frac{1}{80} (16b^4 + 16ab^3 + 16a^2b^2 + 16a^3b + 16a^4 - 40a^4 - 80a^2b^2 - 80a^3b - 80ab^3 - 40b^4 + 40a^4 + 160a^2b^2 + 120a^3b + 120ab^3 + 40b^4 - 15a^4 - 60a^3b - 90a^2b^2 - 60ab^3 - 15b^4)$$

$$= \frac{1}{80} (b^4 - 4a^3b + 6a^2b^2 - 4ab^3 + a^4)$$

$$\mu_4 = \frac{(b-a)^4}{80}$$

Question

Show that for rectangular distribution

$$f(x) = \frac{1}{2a} \quad -a < x < a$$

The m.g.f. about origin is $\frac{1}{at}$ (in hat),

Also show that the moments of even order are given by

$$\mu_{2n} = \frac{a^{2n}}{(2n+1)}$$

Sol

By definition m.g.f.

$$M(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} \cdot f(x) dx$$

$$= \frac{1}{2a} \int_{-a}^a e^{tx} dx$$

$$= \frac{1}{2a} \left[\frac{e^{tx}}{t} \right]_{-a}^a$$

$$M_0(t) = \frac{1}{2a} \left(\frac{e^{at} - e^{-at}}{t} \right)$$

$$= \frac{1}{at} \left(\frac{e^{at} - e^{-at}}{2} \right)$$

$$\underline{M_0(t) = \frac{1}{at} \sinh(at)}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

using the series of $\sinh x$,

$$M_0(t) = \frac{1}{at} \left[at + \frac{(at)^3}{3!} + \frac{(at)^5}{5!} + \dots \right]$$

$$M_0(t) = 1 + \frac{(at)^2}{3!} + \frac{(at)^4}{5!} + \dots$$

Since there is no term of odd power in $M_0(t)$.
So all moments of odd order about origin vanishes
i.e.

$$\mu'_{2n+1} = 0$$

$$\Rightarrow \mu_{2n+1} = 0$$

the moments of even order are given by $\sinh$
Comparing coefficients of $\frac{t^{2n}}{(2n)!}$

$$\mu'_{2n} = \frac{a^{2n}}{(2n+1)} \quad \text{put } n=1$$

$$\mu'_2 = \frac{a^2}{3}$$

$$\text{As } \mu'_x = \mu_x$$

$$n=2$$

$$\mu'_4 = \frac{a^4}{5}$$

$$\text{Therefore } \mu_{2n} = \frac{a^{2n}}{2n+1}$$

Note when mean is zero the moments about origin and moments about mean must be equal $\mu_x = \mu_r$